

Appendix: The Logarithmic Map Separates Scale and Quantization

— g is Factored Out as a Privileged Scale, ω Remains Scale-Invariant

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Purpose

This appendix develops independently the core of the related paper, §7.3 (“The heart of the map: separation of scale (g) and quantization (ω)”). It adds no new claim; its aim is to make explicit the most easily misread point — that **flattening by the logarithmic map is not “throwing away information” but separating scale (g) and quantization (ω).**

Ordinarily, flattening in a symmetric limit is read as “structure vanishes / becomes trivial.” But the logarithmic map maps both v and δ by the same map; it **factors the scale (g) out as a privileged, separable coordinate**, while ω (Δ) remains invariant because Δ is scale-invariant. This is the separation. That the physics (quantization) does not vanish is because quantization resided from the start not in g (scale) but in ω (the fluctuation, area — a scale-invariant half-bit). This holds in any dimension and requires no full symmetry.

This appendix is not a rigorous proof but an explication of the structure of §7.3 of the parent paper. The magnitude of the fluctuation $\pm 1/2$ is a posit (parent paper §6.2). No physical identification is made. “Separation” here is meant only operationally, and includes no claim of a canonical transformation strictly preserving the symplectic form (its rigorization is future work).

1. The two quantities: g and ω

This system has two independent real conserved quantities (parent paper, Axioms 4 and 5; and the appendix “The Complex Structure is a Consequence, Not a Posit”). The two correspond to the real and imaginary parts of the complex inner product $\langle \cdot, \cdot \rangle = g + i\omega$.

g (real part, distance, scale): the sum of squares

$$g = \sum \nu_n^2 = R^2.$$

This appears to the observer as the scale R (the size, distance, metric of the system), the side of normalization (the Born norm). It is the invariant of $SO(2)$ — the rotation preserving the sum of squares.

ω (imaginary part, area, quantization): the product of fluctuations

$$\omega = \delta_1 \delta_2 = \frac{1}{2}.$$

This is an area / symplectic form, the side carrying quantization. It is related to $SO(1,1)$ — the hyperbolic, product-preserving transformation.

By the note of the parent paper, g and ω are independent quantities, mutually underivable. One carries scale (a continuous magnitude), the other quantization (the discrete minimal unit).

2. The logarithmic map separates g

Introduce the fundamental quantity as the logarithm of a squared quantity, $N = \log_2 v$ (parent paper §4). This map **separates the scale (g) side as a privileged coordinate**.

The product (conjugate) relation $v \lambda = k$ becomes, under the logarithm, the additive identity $\sum N_i = K$. **Canonicalization:** separate the conserved quantity as a privileged scale $C = k^{1/n}$ and take the normalized shape $N_i = \log_2(v_i/C)$ ($\sum N_i = 0$). This **holds in any dimension and requires no full symmetry** (the exact formalization of the forward / inverse maps is in the appendix “The maps $(v, \delta) \leftrightarrow (N, \Delta)$ ”). The source of scale is the square (power), and the logarithm linearizes the power.

g can be factored out as a gauge. The scale C is the sole coordinate passed through unchanged; separating it out (or discarding it) leaves a scale-invariant shape. No structure carrying quantization remains on the g side — because g carried scale and never carried quantization.

3. The fluctuation δ is mapped the same way, but Δ is scale-invariant

The fluctuation δ , too, is mapped by the **same map** $\text{func} = \log_2$ as the central value v : $\delta \rightarrow \Delta = \log_2(\text{the fluctuation factor})$ (appendix “The maps $(v, \delta) \leftrightarrow (N, \Delta)$,” §4). That is, **the logarithmic map does act on δ** — this is a canonicalization of the same form as $v \rightarrow N$ and must not be ignored.

However, the resulting $\Delta = \pm 1/2$ (the half-bit) is **scale-invariant**: independent of the privileged scale C , it is the same $\pm 1/2$ at any scale. Hence, after factoring out C as a gauge, Δ remains as an invariant. **This is what the invariance of ω really is** — not “the logarithmic map does not act on ω ,” but “ δ is mapped the same way to Δ , yet Δ is scale-invariant.”

Concretely, the half-bit $\pm 1/2$ is independent of the scale C and is unchanged when C is separated. The degree of freedom of which member of the conjugate pair the half-bit is allocated to ($\Delta_a + \Delta_b = 0$, sum-zero; appendix “From the Sum-Zero of Fluctuations to Uncertainty”) is also preserved independently of the separation of C .

Δ (ω) remains as the scale-invariant minimal unit of quantization. $\omega = \text{area } \delta_1 \delta_2 = 1/2$ is this scale-invariant half-bit unit, and remains as the minimal unit of quantization even after the scale is separated (gauged away).

4. The structure of the separation

From the above, the logarithmic map separates the two quantities.

Quantity	What it carries	Under the logarithmic map
g (sum of squares, distance, real part)	scale (continuous magnitude)	flattened (removable as a gauge)
ω (product, area, imaginary part)	quantization (discrete minimal unit)	remains invariant

The operational meaning of “separation”: “to separate” means that the map maps both v and δ by the same map, while it factors the scale g (real part) out as a privileged scale C , and ω (Δ , imaginary part) remains as an invariant because Δ is scale-invariant. g and ω are the real and imaginary parts of the complex inner product $g + i\omega$; the map carries both, but the scale (real part) is separable as a gauge, while ω (imaginary part) remains scale-invariant (Figure 1).

The substance of “the physics does not vanish even when flattened”: factoring the scale (g) out as a privileged coordinate, quantization (ω) nonetheless remains because quantization resided from the start not in g (scale) but in ω (the fluctuation, area). The map separates g but does not fold ω . Hence, even when scale is factored out, quantization remains.

This separation is a consequence of quantization (ω) being a conserved quantity independent of, and distinct from, scale (g). Scale is pushed into the flat surface and gauged away, and quantization is purified as an invariant on that flat surface — this is the work of the logarithmic map.

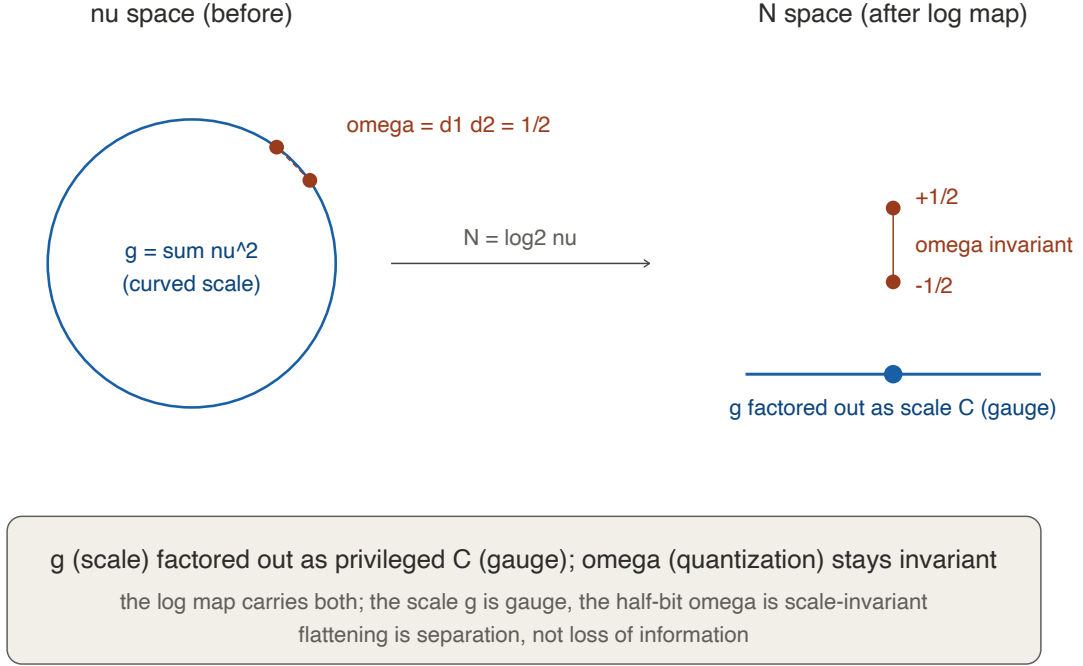


Figure 1: g/ω separation by the logarithmic map. The scale g in ν -space is factored out as a privileged scale C (gauge) under the logarithm. On the other hand, the quantization ω (the half-bit $\pm 1/2$) remains invariant under the map. This is not a loss of information but a separation of g and ω (no full symmetry required).

5. Summary of the structure

For the complex inner product $g+i \omega$, the logarithmic map $N=\log_2 \nu$:

- separates / factors out the real part g (scale) as a privileged coordinate (a gauge; no full symmetry required);
- maps the imaginary part ω (product, quantization, Δ) by the same map, yet because Δ is scale-invariant it remains as an invariant on the flat surface.

Flattening is not a loss of information but a separation of two quantities. Scale (g) carries a continuous magnitude and is flattenable; quantization (ω) carries the discrete minimal unit and is invariant under the map. That the two are independent (the parent paper's note: the mutual underivability of Axioms 4 and 5) is the ground for this separation.

This is the heart of the parent paper — even when flattened in a high-symmetry limit, quantization does not vanish, because quantization resides not in the side that is flattened (g , scale) but in the side that remains invariant (ω , area).

6. Limitations (reading notes)

This appendix makes explicit the structure of §7.3 of the parent paper and adds no new claim.

1. **“Separation” is meant only operationally.** It means the map flattens g and keeps ω invariant. It includes no claim of a canonical transformation strictly preserving the symplectic form. That rigorization (a proof that the map is a canonical transformation) is future work.
2. **$\pm 1/2$ is a posit.** As in §6.2 of the parent paper, the minimal unit of ω , $\pm 1/2$, is a posit, not a derivation, and the separation of this appendix presupposes this posit.
3. **The difference from known frameworks is uncomared.** Whether the g/ω separation of this appendix differs from or agrees with known Kähler geometry (the structure in which a metric g and a symplectic form ω are tied by a complex structure) or conformal flatness has not been rigorously compared. As in the limitation §8.4 of the parent paper, whether this result remains a special case of known frameworks cannot be determined.
4. **Holds in any dimension and for any k .** As in the revised §8.1 of the parent paper, the flattening of the additive structure and the full invertibility of the map (canonicalization: privileged-scale separation) require no full symmetry. The g/ω separation of this appendix likewise holds in any dimension. What lies outside the scope is the extension to a curvature tensor and the linearization of the sum of squares Σv^2 itself.
5. **No physical identification is made.** The physical names of g (scale) and ω (quantization) are anonymous labels, and this appendix makes no identification with a particular physical quantity.

Reference

- Kihara, N. “On the Logarithmic Map of Scale and the Separation of Scale and Quantization in a High-Symmetry Limit — A Limited Verification Report from a Minimal Axiom System,” §7.2, §7.3, §8.1, §8.4, Axioms 4 and 5. Concept DOI: [10.5281/zenodo.20740841](https://doi.org/10.5281/zenodo.20740841).